8 *Use of the Data Booklet is relevant to this question.*

A student mixes 20.0 cm^3 of 5.00 mol dm^{-3} sulfuric acid with an equal volume of 6.00 mol dm^{-3} sodium hydroxide. The initial temperature of both solutions is 25.0 °C. The maximum temperature reached after the reaction is 55.0 °C. Assume the density of both solutions is 1 g cm^{-3}.

What is the value of the enthalpy change of neutralisation, in kJ mol^{-1}, calculated using these values?

A –41.8

B –50.2

C –83.6

D –100.3

Commented [SXF(5]: Neutralisation and I included the units cos the A level qns has the units in the opions